Satellite Imagery-Based Property Valuation

1. Overview

This project focuses on predicting residential house prices by combining **structured tabular property data** with **satellite imagery**, with the goal of capturing both intrinsic property characteristics and surrounding environmental context.
Dataset Description

- **Total properties available**: ~16,000
- **Training set**: ~13,000 properties
- **Validation set**: ~2,800 properties
- **Test set**: ~5,400 properties
- **Satellite image coverage**: 100% for training and validation data

Each property is associated with geographic coordinates, enabling retrieval of corresponding satellite images.

Given Data
The provided dataset contains structured tabular information, including:

- Living area, number of bedrooms and bathrooms
- Construction quality and year built
- Geographic coordinates (latitude, longitude)
- Transaction price (modeled as log-transformed price)

This tabular data serves as the **primary baseline** for price prediction.

External Data via API

Satellite images are retrieved using the **Sentinel Hub API** (Sentinel-2 Level-2A imagery).

For each property:

- Latitude and longitude are treated as the **center point**
- A fixed-size bounding box is generated around the center
- RGB satellite images are fetched within a defined time window
- Cloud coverage filtering is applied for image quality

The images capture **greenery, urban density, road networks, and open spaces**, which are not explicitly present in tabular data.

Data Processing & Modeling Strategy

Two parallel pipelines are used:

Tabular Pipeline

- Removal of target-related and non-numeric columns to avoid leakage
- Feature scaling using StandardScaler

- Target modeled as log(price)

Image Pipeline

- Images resized to CNN-compatible resolution
- Pretrained CNN (ResNet) used as a **frozen feature extractor**
- Image embeddings generated for each property

Two models are evaluated:

- **Tabular-only model**
- **Tabular + Satellite (Fusion) model**

All evaluation is performed on **held-out test data**.

Objective

The primary objective of this project is to assess whether **satellite imagery provides additional predictive and economic value** beyond traditional tabular data. By comparing the performance of the two models, the project quantifies the contribution of visual environmental context in house price estimation.

Output

The final outputs of the project include:

- Predicted house prices for unseen properties.
- A quantitative comparison between tabular-only and multimodal models.
- Insights into how environmental visual features influence property valuation.

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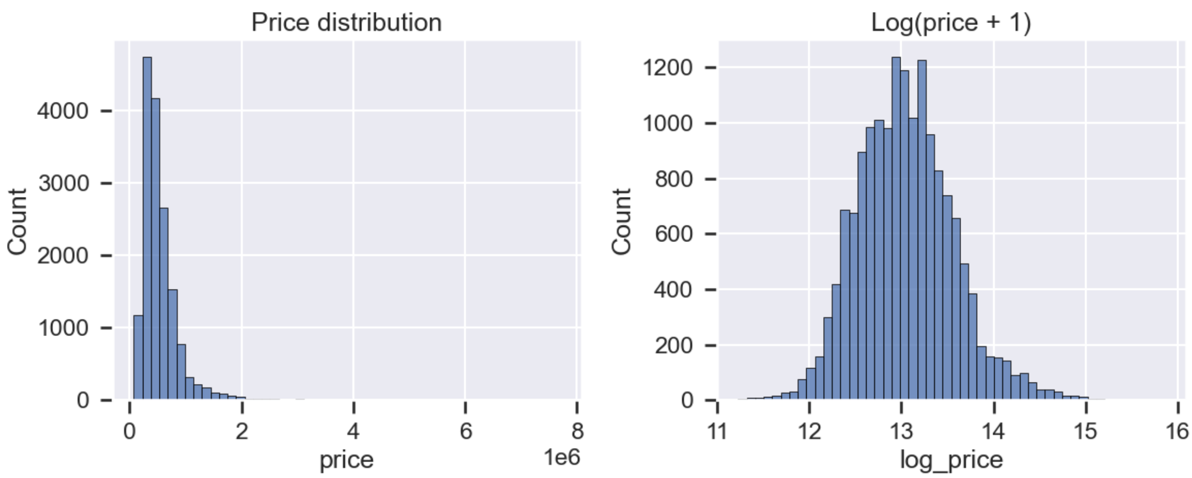
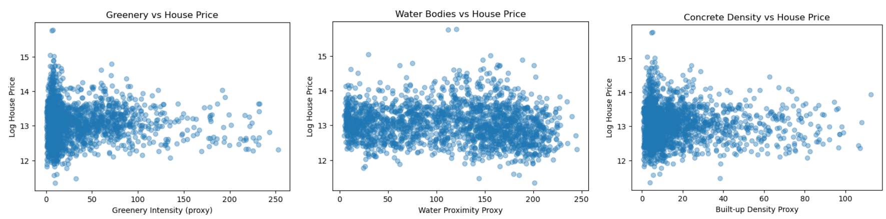
2. EDA

Price Distribution Analysis



- House prices exhibit a strong **right-skew**
- Log-transformation produces a more symmetric distribution
- This stabilizes variance and improves regression performance



Sample Satellite Image Visualization

Sample satellite images show large variation across properties:

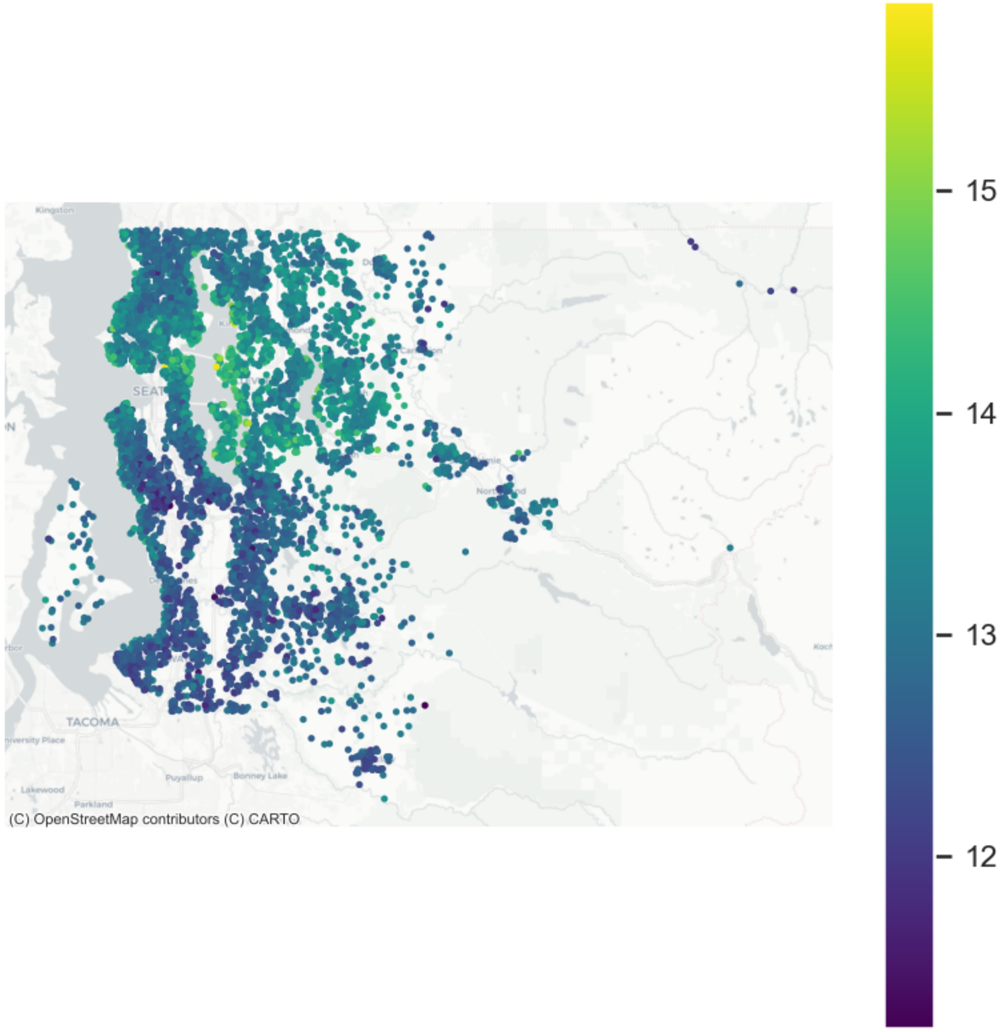
- Green and open residential areas
- Dense concrete and congested urban regions

These variations motivate the inclusion of satellite imagery as an additional modality.

EDA Summary

- Structural features strongly influence price
- Spatial and environmental differences are visually evident
- Satellite imagery captures neighborhood context not fully encoded in tabular features

Geospatial Distribution of Properties



3. Financial / Visual Insights

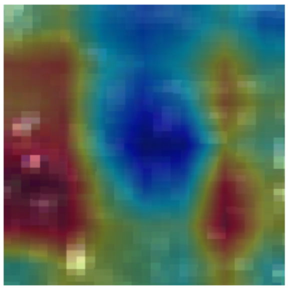
High-Priced Property Features (Positive Value Drivers)

- **Green Cover and Vegetation**
  High-priced properties are commonly located in areas with **visible green cover**, such as trees, parks, and open vegetation. These surroundings improve air quality, reduce noise, and enhance aesthetic appeal, making the neighborhood more desirable for residents

- **Open Spaces and Low Density**
  Premium properties often exhibit **open layouts** with sufficient spacing between structures. Such environments provide better sunlight, ventilation, and a sense of privacy. The model consistently associates these open spatial patterns with higher predicted prices.

- **Planned Layout and Accessibility**
  High-value neighborhoods typically show **organized road networks and planned layouts**, reflecting better infrastructure and connectivity. These features signal ease of access to amenities and long-term livability, contributing to higher property values.

Satellite image of
high priced property



Grad-CAM of
high priced property

Grad-CAM visualizations show strong model attention on **green and open regions**, indicating positive valuation signals.

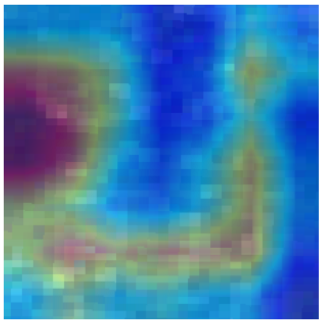
Low-Priced Property Features (Negative Value Drivers)

- **Concrete Density and Urban Congestion**
  Low-priced properties are frequently surrounded by **dense concrete structures** with tightly packed buildings and minimal greenery. These environments indicate congestion, higher noise levels, and limited living space. Grad-CAM highlights these dense built-up regions as important negative cues, showing that the model learns concrete density as a **price-reducing signal**.

- **Limited Open Space**
  A lack of visible open areas is a common characteristic of lower-valued properties. Such regions often feel overcrowded and less livable, which negatively affects buyer perception and market value.

- **Chaotic Road Patterns**
  Unstructured or narrow road networks often appear in lower-priced areas, suggesting poor accessibility and infrastructure stress. The model captures these visual patterns and associates them with reduced property valuations.

Satellite image of
low priced property



Grad-CAM of
low priced property

Grad-CAM highlights dense built-up areas as **negative price signals**, confirming that the model learns meaningful visual cues.

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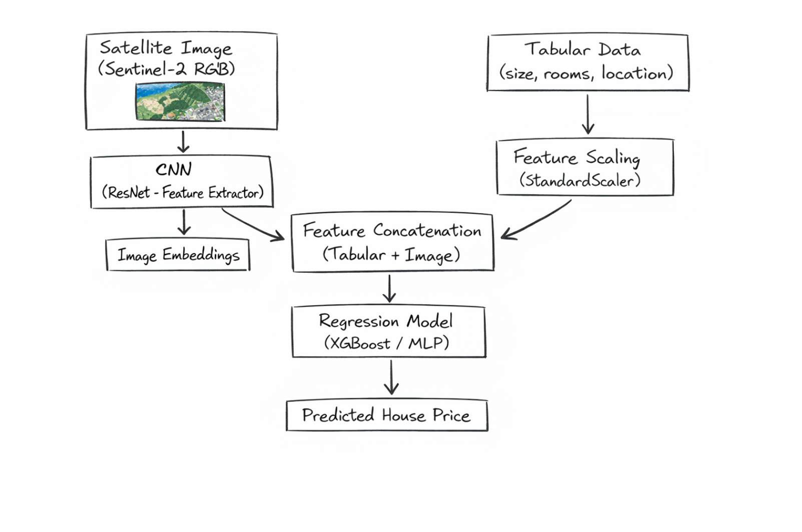
4. Architecture Diagram



Satellite images are processed using a CNN to extract environmental features, which are concatenated with scaled tabular features and passed to a regression model.

5. Results

Model	Test R^2 (log)	Test RMSE (log)	Test RMSE (₹)

Tabular Data Only(XGBoost)	**0.895**	**0.1674**	**₹104,337**
Tabular + Satellite (Fusion XGBoost)	0.867	0.1941	₹128,593

Observations

- The **Tabular-only model** achieves the best numerical performance, indicating that structural and location-based features already explain a large portion of house price variance.
- The **fusion model** shows a slightly lower R^2 and higher RMSE, suggesting that adding satellite imagery introduces additional noise along with useful signal.
- Despite lower aggregate metrics, the multimodal model captures **environmental and neighbourhood-level information** that is not present in tabular data alone.

Why the Fusion Model Is Still Important

Although the tabular model performs better numerically, the fusion model is **conceptually stronger**:

- Satellite images encode **greenery, open spaces, urban density, and layout**, which are economically meaningful but difficult to represent explicitly.
- Grad-CAM analysis confirms that the CNN focuses on **relevant visual regions** (trees, open areas, dense concrete), indicating that the visual branch learns non-random patterns.
- The slightly lower performance highlights the challenge of effectively integrating noisy real-world imagery, rather than the absence of useful visual information.

For this project, the **fusion model is selected** to demonstrate the feasibility and interpretability of **multimodal learning**, even when gains are not purely metric-driven.

Model Selection

Although the tabular-only model performs best in terms of RMSE and R^2, the **fusion model is selected** for this project to demonstrate:

- The feasibility of multimodal learning
- The interpretability of visual features via Grad-CAM
- The economic relevance of environmental context

Conclusion

This project demonstrates that while tabular features remain the strongest predictors of house prices, satellite imagery provides **meaningful contextual information** related to greenery, density, and neighborhood structure. Although the multimodal model does not outperform the tabular baseline numerically, visual explainability confirms that satellite images encode economically relevant signals. The study highlights both the **promise and challenges** of multimodal learning for real-world real estate valuation.

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